SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS

CERTIFICATE OF INSTALLATION

Note: This table completed by ~~HERS~~ECC Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

Note: This Certificate of Installation document is only applicable if a PV system is in the proposed design or PV Exceptions are selected in the CF1R-PRF-01 or CF1R-NCB-01.

A. General Information

01	Project Location (City)		02	Building Type	
03	Climate Zone		04	Method of Compliance:	
05	Qualifying Exceptions		06	Community Solar	

B. Design Photovoltaic Systems Information

01	02	03	04	05	06	07	08	09	10	11	12	13
PV Array ID or Name	~~Adjusted Minimum~~Adjusted Minimum PV Size (kW)	Adjusted Minimum PV Size from qualified exception requirement~~Value from Exception~~	Module Type	CFI (Yes/No)	Azimuth (deg)	Tilt Input (Deg/Pitch)	Angle /Tilt	Annual Solar Access (%)	Inverter Efficiency (%)	Shading Requirement Compliance Path	Array Type	Module Level Power Electronics
14	Total DC System Size (kW)											

C. Installed Photovoltaic Systems Information

If the installer certifies that the installed PV system matches or exceeds the design PV system, the building complies with the PV system requirement, otherwise it does not comply.

01	02	03	04	05	06	07	08	09	10
PV Array ID or Name	DC System Size (kW)	Module Type	Azimuth (deg)	Tilt Input (Deg/Pitch)	Angle/Tilt	Annual Solar Access (%)	Inverter Efficiency (%)	Array Type	Module Level Power Electronics
11	Total DC System Size (kW)								

D. Solar Access Verification

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The installer shall provide documentation that demonstrates the shading condition of the actual installation of the PV module is consistent with the annual solar access % in Table B. The verification must be done by measurements from an approved solar assessment tool or other CEC approved alternative methods. The satellite, drone or other digital image of the obstructions that cast shadows on the PV array must be created and dated after the installation of the photovoltaic system. If the image is dated before the installation, then additional on-site pictures must be attached to clearly show that the installed system matches the system modeled in the solar assessment report.

E. System Monitoring Requirements

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

All installed PV system must have a working web based portal and a mobile device application provide access to the following information

01	Nominal kW rating of the PV system
02	Number of PV modules and nominal watt rating of each module
03	Hourly (or 15 min), daily, monthly and annual kWh production in numeric and graphic format
04	Running total of daily kWh production
05	Daily kW peak power production
06	Current kW production of the entire PV system

F. Qualifying Exception Requirement

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

Limited Solar Access
The installer shall provide documentation of the solar access roof area (SARA) limitations that justify the exception. Documentation may include roof plans, aerial photos, satellite images, 3D model, or other documentation that clearly shows the available solar access roof areas that meet~~s~~ the definition of SARA in 150.1(c)14~~solar access requirements.~~
Declared emergency area
If a building is damaged or destroyed in a declared emergency area prior to 1/1/2020 (AB-178), it must comply with PV requirement applicable on originally constructed permit date. Eligibility to this exception, such as income or insurance requirements, shall be confirmed by the enforcement agency.
Snow Load
The installer shall provide roof design, PV system design, and/or ASCE Standard 7-~~16~~22, Chapter 7, Snow Loads calculation to the enforcement authority. The enforcement authority must determine that it is not possible for the PV system, including panels, modules, components, supports, and attachments to the roof, to meet ASCE Standard 7-~~16~~22, Chapter 7, Snow Loads.
10-109(k) PV Requirement Determination
Only buildings within the jurisdiction of Trinity Public Utility District ~~or the City of Needles~~ qualify for this exception.

G. SMUD Solar Share Program

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Required kW	
02	Standard Design kW (CFI1 orientation and 98% solar access)	
03	Attach a copy of SMUD Attestation of Premise Registration in Neighborhood SolarShares (Attestation).	

H. Compliance Statement

DOCUMENTATION AUTHOR'S DECLARATION STATEMENT

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/~~HERS~~ECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-PVB-01-E User Instructions

A. General Information

1. For information only and requires no user input.
2. For information only and requires no user input.
3. User choose from list of qualifying exceptions to the PV requirements. If no exception applicable, choose N/A
4. For information only and requires no user input.
5. For information only and requires no user input.
6. For information only and requires no user input.

B. Design Photovoltaic Systems Information

This table reports the PV system features that were specified on the registered CF1R compliance document for this project. For information only and requires no user input.

C. Installed Photovoltaic Systems Information

1. PV Array ID or Name - Reference information from CF1R.
2. DC System Size – Enter the kWdc of the array. Must be equal or greater the adjusted design system size for this array.
3. Module Type – If the array meets the California Flexible Installation criteria, then enter the Module Type. Different module types are Standard and Premium.
4. Azimuth - If the array meets the California Flexible Installation criteria, then enter the azimuth of the array in degrees from North.
5. Tilt Input - Different Tilt input are Degree and Pitch.
6. Angle/Tilt - Enter the value of the angle or tilt.
7. Annual Solar access – Enter the percent of solar access
8. Inverter Efficiency – Enter the inverter efficiency in percent. Must be equal or greater the design inverter efficiency for this array.
9. Array Type – Choose from: fixed (open rack), tracking (one axis), tracking (two axis)
10. Module Level Power Electronics – Choose from: microinverters or DC power optimizers

D. Solar Access Verification

Installer must ensure all the requirements on this table are met.

E. System Monitoring Requirements

Installer must ensure all the requirements on this table are met.

F. Qualifying Exception Verification

Installer must ensure all the requirements on this table are met.

G. SMUD Solar Share Program

Installer must ensure all the requirements on this table are met.

Note: This Certificate of Installation document is only applicable if a PV system is in the proposed design or PV Exceptions are selected in the CF1R-PRF-01 or CF1R-NCB-01

A. General Information					
01	Project Location (City)	<<Auto filled field text: Reference text from CF1R>>	02	Building Type	<<Auto filled field text: Reference text from CF1R >>
03	Climate Zone	<<Auto filled field text: Reference text from CF1R >>	04	Method of Compliance:	<<Reference CF1R document: allowed values: Performance or Prescriptive>>
05	Qualifying Exceptions	<< If CF1R-PRF report Battery storage or Community solar then A05= NA Else user pick from list: Limited Solar Access No PV – SARA less than 80 contiguous sq ft (Trigger CF2R-SRA-01) No PV – required PV less than 1.8 kWdc Plan approved before 1/1/20 Battery storage ~~(Trigger CF2R-PVB-02)~~ Community Solar Declared emergency area before 1/1/20 No PV – Snow loads Section 10-109(k) PV determination NA >>	06	Community Solar	<<Auto filled field text: Reference text from CF1R; allow NA>>

B. Design Photovoltaic Systems Information
<<if A05 = "No PV – SARA less than 80 contiguous sq ft ", "No PV – required PV less than 1.8 kWdc", "Community Solar", "Declared emergency area before 1/1/20", "No PV – Snow loads", or "No PV - Section 10-109(k) PV determination", then display the "section does not apply" message; else display this entire table >>

01	02	03	04	05	06	07	08	09	10	11	12	13
PV Array ID or Name	~~Adjusted Minimum~~ Adjusted Minimum PV Size (kW)	Adjusted Minimum PV Size from qualified exception requirement (kW)~~Value from Exception~~	Module Type	CFI	Azimuth (deg)	Tilt Input (Deg/Pitch)	Angle/Tilt	Annual Solar Access (%)	Inverter Efficiency (%)	Shading Requirement Compliance Path	Array Type	Module Level Power Electronics
<<auto filled text: referenced from CF1R; elseif not available, user input>>	<<auto filled text: referenced from CF1R>>	<<if performance, then value = B02; Elseif prescriptive and A05 = NA, then ~~autofill from B02; elseif A05 = "Battery storage", then value = ((O04 from CF1R*0.75) + O05~~	<<From CF1R-PRF-01; Else = NA>>	<<From CF1R-PRF-01 (Da06_CFI_PV), value = Yes (true) if CFI1 or CFI2 is chosen, else value	<<If performance and CFI = Yes, then if CF1R-PRF Da07_AzimuthRange = 150 to 270, user input between 150 and 270; elseif CF1R-PRF Da07_AzimuthRange = 105 to 300, then user input between 105 and 300; else if	<<From CF1R-PRF-01; else user pick from list: ☐ Deg ☐ Pitch >>	<<If prescriptive and B07=Deg and (B06≤59 or B06≥301), then user input 0 ≤B08≤ 10; elseif prescriptive and B07=Deg and 90≤B06≤300, then user input 0 ≤B08<90; elseif prescriptive and B07=Pitch and (B06≤59 or B06≥301),	<<From CF1R-PRF-01; Else value = 98>>	<<From CF1R-PRF-01; Else = NA>>	<<Default value = "Minimum Shading Criterion">>	<<From CF1R-PRF-01; Else = NA>>	<<From CF1R-PRF-01; Else = NA>>

		~~from CF1R);~~O09 from CF1R Else user input: decimalnonnegative >>		= No (false); Else if prescriptive, value = NA>>	performance and CFI = Yes and value is not on CF1R-PRF, then user input between 150 and 270; else if performance and CFI=No, use value from CF1R-PRF Da07_Azimuth; else if performance and CFI=No and value is not on CF1R-PRF, then user input between 0 and 359; if prescriptive, then user input between 0 and 359>>		then user input 0 ≤B08≤ 2; elseif prescriptive and B07=Pitch and 90≤B06≤300, then user input 0 ≤B08≤50; elseif performance and CFI = Yes, then value from CF1R-PRF and ≤ 7; elseif performance and CFI = No, then value from CF1R-PRF>>					
14	Total DC System Size (kW)			<<Sum of B03>>								

C. Installed Photovoltaic Systems Information

<<if A05 = “No PV – SARA less than 80 contiguous sq ft”, “No PV – required PV less than 1.8 kWdc”, “Community Solar”, “Declared emergency area before 1/1/20”, “No PV – Snow loads”, or “No PV - Section 10-109(k) PV determination”, then display the "section does not apply" message; else display this entire table>>

<<For each record in table B, require a record in Table C. If B05 is Yes for an array, allow user to add a record in Table C for that array.

01	02	03	04	05	06	07	08	09	10
PV Array ID or Name	DC System Size (kW)	Module Type	Azimuth (deg)	Tilt Input (Deg/Pitch)	Angle/Tilt	Annual Solar Access (%)	Inverter Efficiency (%)	Array Type	Module Level Power Electronics
<<auto filled text: referenced from CF1R; else user input>>	<<If B05=No, then autofill from B03 but allow user to override; Else user input >>	<<If B05=No, then autofill from B04; Else user pick from list: Standard, Premium >>	<<If B05=No, then autofill from B06; elseif B05 = Yes, then if CF1R-PRF Da07_Azimuth Range = 150 to 270, user input between 150 and 270; Elseif CF1R-PRF Da07_Azimuth Range = 105 to 300, then user input between 105 and 300; Else user input (value must be > 0 and ≤ 359>>	<<If B05=No, then autofill from B07; Else user pick from list: ☐ Deg ☐ Pitch>>	<<If B05=No, then autofill from B08; Else user input (value must be > 0 and < 10)>>	<<reference value from B09 as default, but allow user to override >>	<<reference value from B10 as default, but allow user to override with a higher number>>	<<reference value from B12>>	<<reference value from B13 as default, if B13 = “none”, then allow user to override and pick from list: *Microinverters or *DC power Optimizers>>
11	Total DC System Size (kW)	<<Sum of C02>>							

If the installer certifies that the installed PV system matches or exceeds the design PV system, the building complies with the PV system requirement, otherwise it does not comply.

D. Solar Access Verification

<<if A05 = “No PV – SARA less than 80 contiguous sq ft”, “No PV – required PV less than 1.8 kWdc”,“Community Solar” , “Declared emergency area before 1/1/20”, “No PV – Snow loads”, or “No PV - Section 10-109(k) PV determination”, then display the "section does not apply" message; else display this entire table >>

01	The installer shall provide documentation that demonstrates the shading condition of the actual installation of the PV module is consistent with the annual solar access % in Table B. The verification must be done by measurements from an approved solar assessment tool or other CEC approved alternative methods. The satellite, drone or other digital image of the obstructions that cast shadows on the PV array must be created and dated after the installation of the photovoltaic system. If the image is dated before the installation, then additional on-site pictures must be attached to clearly show that the installed system matches the system modeled in the solar assessment report.

The responsible person’s signature on this compliance document affirms that all applicable requirements in this table have been met.

E. System Monitoring Requirements

<<if A05 = “No PV – SARA less than 80 contiguous sq ft ”, “No PV – required PV less than 1.8 kWdc,“Community Solar” , “Declared emergency area before 1/1/20”, “No PV – Snow loads”, or “No PV - Section 10-109(k) PV determination”, then display the "section does not apply" message; else display this entire table >>

All installed PV system must have a working web based portal and a mobile device application provide access to the following information:

01	Nominal kW rating of the PV system
02	Number of PV modules and nominal watt rating of each module
03	Hourly (or 15 min), daily, monthly and annual kWh production in numeric and graphic format
04	Running total of daily kWh production

05	Daily kW peak power production
06	Current kW production of the entire PV system
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

F. Qualifying Exception Requirement
<<If A05 "Qualifying Exceptions" = "NA" or "Community Solar", then display the "section does not apply" message; Else If A05 "Qualifying Exceptions" = "No PV – SARA less than 80 contiguous sq ft", , or "Plan approved before 1/1/20" then display row "Limited Solar Access" below; Else if A05 "Qualifying Exceptions" = "Declared emergency area before 1/1/20" then display row "Declared emergency area" below; Else if A05 "Qualifying Exceptions" = No PV – Snow Load" then display row "Snow load" below; Else if A05 "Qualifying Exceptions" = "Section 10-109(k) PV determination" then display row "Section 10-109(k) PV Requirement Determination">>
Limited Solar Access
The installer shall provide documentation of solar access the roof area (SARA) limitations that justify the exception. Documentation may include roof plans, aerial photos, satellite images, 3D model, or other documentation that clearly shows the available solar access roof areas that meet~~s~~ the definition of SARA in 150.1(c)14 ~~solar access requirements~~.
Declared emergency area
If a building is damaged or destroyed in a declared emergency area prior to 1/1/2020 (AB-178), it must comply with PV requirement applicable on originally constructed permit date. Eligibility for this exception, such as income or insurance requirements, shall be confirmed by the enforcement agency.
Snow Load
The installer shall provide roof design, PV system design, and/or ASCE Standard 7-~~16~~22, Chapter 7, Snow Loads calculation to the enforcement authority. The enforcement authority must determine that it is not possible for the PV system, including panels, modules, components, supports, and attachments to the roof, to meet ASCE Standard 7-~~16~~22, Chapter 7, Snow Loads.
10-109(k) PV Requirement Determination
Only buildings within the jurisdiction of Trinity Public Utility District ~~or the City of Needles~~ qualify for this exception.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

G. SMUD Solar Share Program <<If A05 ≠ "Community Solar", then display the "section does not apply" message; else display this entire table >>		
01	Required kW	<< From CF1R-PRF-01>>
02	Standard Design kW (CFI1 orientation and 98% Annual Solar Access)	<< From CF1R-PRF-01>>
03	Attach a copy of SMUD Attestation of Premise Registration in Neighborhood SolarShares (Attestation).	
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.		

H. Compliance Statement
<<calculated field: if C11 ≥ B14 or A05 = "No PV – SARA less than 80 contiguous sq ft", "No PV – required PV less than 1.8 kWdc", or "Community Solar", then display result: Pass - ~~dwelling~~ single family home complies with the Photovoltaic Systems requirements; else display result: Fail - ~~dwelling~~ single family home does not comply with the Photovoltaic System requirements>>